# PG No. 14 $D_{2\mathrm{d}}$ $\bar{4}2m$ (-42m setting) [ tetragonal ]

Table 1: Representation matrices

| Irrep. | | | | | | |
|---|---|---|---|---|---|---|
| $A_1$ | $1:\begin{bmatrix}1\end{bmatrix}$ | $2_{001}:\begin{bmatrix}1\end{bmatrix}$ | $-4^+_{001}:\begin{bmatrix}1\end{bmatrix}$ | $-4^-_{001}:\begin{bmatrix}1\end{bmatrix}$ | $2_{010}:\begin{bmatrix}1\end{bmatrix}$ | $2_{100}:\begin{bmatrix}1\end{bmatrix}$ |
| | $\mathrm{m}_{110}:\begin{bmatrix}1\end{bmatrix}$ | $\mathrm{m}_{1-10}:\begin{bmatrix}1\end{bmatrix}$ | | | | |
| $A_2$ | $1:\begin{bmatrix}1\end{bmatrix}$ | $2_{001}:\begin{bmatrix}1\end{bmatrix}$ | $-4^+_{001}:\begin{bmatrix}1\end{bmatrix}$ | $-4^-_{001}:\begin{bmatrix}1\end{bmatrix}$ | $2_{010}:\begin{bmatrix}-1\end{bmatrix}$ | $2_{100}:\begin{bmatrix}-1\end{bmatrix}$ |
| | $\mathrm{m}_{110}:\begin{bmatrix}-1\end{bmatrix}$ | $\mathrm{m}_{1-10}:\begin{bmatrix}-1\end{bmatrix}$ | | | | |
| $B_1$ | $1:\begin{bmatrix}1\end{bmatrix}$ | $2_{001}:\begin{bmatrix}1\end{bmatrix}$ | $-4^+_{001}:\begin{bmatrix}-1\end{bmatrix}$ | $-4^-_{001}:\begin{bmatrix}-1\end{bmatrix}$ | $2_{010}:\begin{bmatrix}1\end{bmatrix}$ | $2_{100}:\begin{bmatrix}1\end{bmatrix}$ |
| | $\mathrm{m}_{110}:\begin{bmatrix}-1\end{bmatrix}$ | $\mathrm{m}_{1-10}:\begin{bmatrix}-1\end{bmatrix}$ | | | | |
| $B_2$ | $1:\begin{bmatrix}1\end{bmatrix}$ | $2_{001}:\begin{bmatrix}1\end{bmatrix}$ | $-4^+_{001}:\begin{bmatrix}-1\end{bmatrix}$ | $-4^-_{001}:\begin{bmatrix}-1\end{bmatrix}$ | $2_{010}:\begin{bmatrix}-1\end{bmatrix}$ | $2_{100}:\begin{bmatrix}-1\end{bmatrix}$ |
| | $\mathrm{m}_{110}:\begin{bmatrix}1\end{bmatrix}$ | $\mathrm{m}_{1-10}:\begin{bmatrix}1\end{bmatrix}$ | | | | |
| $E$ | $1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$ | $2_{001}:\begin{bmatrix}-1&0\\0&-1\end{bmatrix}$ | $-4^+_{001}:\begin{bmatrix}0&1\\-1&0\end{bmatrix}$ | $-4^-_{001}:\begin{bmatrix}0&-1\\1&0\end{bmatrix}$ | $2_{010}:\begin{bmatrix}-1&0\\0&1\end{bmatrix}$ | $2_{100}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$ |
| | $\mathrm{m}_{110}:\begin{bmatrix}0&-1\\-1&0\end{bmatrix}$ | $\mathrm{m}_{1-10}:\begin{bmatrix}0&1\\1&0\end{bmatrix}$ | | | | |